8. P and Q can do a piece of work in 12 days and 15 days respectively. P started the work alone and then after 3 days, Q joined him till the work was completed. How long did the work last?

Challenging Problems

9. If the numerator of a fraction is increased by 50% and the denominator is decreased by 20%, then it becomes $\frac{3}{5}$. Find the original fraction.

10. Gopi sold a laptop at 12% gain. If it had been sold for ₹1200 more, the gain would have been 20%. Find the cost price of the laptop.



11. A shopkeeper gives two successive discounts on an article whose marked price is ₹180 and selling price is ₹108. Find the first discount percentage if the second discount is 25%.

12. Find the rate of compound interest at which a principal becomes 1.69 times itself in 2 years.

13. A small–scale company undertakes an agreement to make 540 motor pumps in 150 days and employs 40 men for the work. After 75 days, the company could make only 180 motor pumps. How many more men should the company employ so that the work is completed on time as per the agreement?

14. P alone can do $\frac{1}{2}$ of a work in 6 days and Q alone can do $\frac{2}{3}$ of the same work in 4 days. In how many days will they finish $\frac{3}{4}$ of the work, working together?

15. X alone can do a piece of work in 6 days and Y alone in 8 days. X and Y undertook the work for ₹48000. With the help of Z, they completed the work in 3 days. How much is Z's share?

SUMMARY

- When the S.P is more than the C.P, then there is a gain or profit. *Profit/Gain* = *S.P – C.P.*
- When the S.P is less than the C.P, then there is a loss. *Loss* = *C.P – S.P.*
- The profit or loss is always calculated on the cost price.
- Selling price = Marked price – Discount.
- **Formulae**

(i) Profit or Gain % $= \left(\frac{Profit}{C.P} \times 100\right)\%$

(ii) Loss % $= \left(\frac{Loss}{C.P} \times 100\right)\%$

(iii) Selling Price, S.P $= \frac{(100 + Profit\%)}{100} \times$ C.P (or) Cost Price, C.P $= \frac{100}{(100 + Profit\%)} \times$ S.P

(iv) Selling Price, S.P $= \frac{(100 - Loss\%)}{100} \times$ C.P (or) Cost Price, C.P $= \frac{100}{(100 - Loss\%)} \times$ S.P

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